

COMPREHENSIVE INDUSTRIAL ENGINEERING & MANUFACTURING OPTIMIZATION REPORT

Wheel Assembly Workstation Cycle Time Analysis, Operational Bottleneck Elimination, and Cellular Facility Layout Reconfiguration

Lead Industrial Engineer: Yash Jain	Roll / Student ID: 25SCS1003003010
Program: Operations Industrial Engineer Job Simulation	Institution / Platform: Siemens Mobility • Forage
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EXECUTIVE SUMMARY & PROJECT CHARTER

Under Siemens Mobility's high-speed rail enhancement charter, **Project Velocity** was commissioned to evaluate manufacturing throughput for the **Aurora Express** bogie manufacturing division. Empirical capacity modeling identified the **Wheel Assembly Section** as the paramount production constraint, establishing an operational bottleneck that limits whole-train manufacturing takt pacing. The assembly department operates at an average cycle time of **62 minutes and 4 seconds per wheelset** across four assembly bays, failing to satisfy the required daily output rate of 16 wheelsets per two-shift operating day.

This technical dossier delivers an end-to-end industrial engineering optimization. **Section I** presents a granular statistical time-motion study across 20 discrete operational work elements over 10 production cycles, isolating micro-variations and equipment starvation patterns. **Section II** investigates the mechanical and structural failure modes of the primary bottleneck operations. **Section III** audits the physical shop floor architecture and details how the legacy layout induces acute overhead crane interlocking and transport waste. **Section IV** introduces an innovative U-shaped cellular layout with decentralized handling, and **Section V** outlines an empirical risk mitigation and deployment roadmap. The officially verified certificate of completion is appended as Section VI.

62:04 min	05:24 min	38:15 min	+62.4%
BASELINE CYCLE TIME	MAX TASK (MOUNT AXLE)	OPTIMIZED CYCLE TARGET	THROUGHPUT EXPANSION

Empirical Workstation Time Study Analysis (Part 1: Tasks 1–10)

A formal time-motion study was conducted on the Aurora Express wheel assembly line in accordance with REFA and Maynard's Industrial Engineering standards. Ten consecutive full manufacturing cycles were observed and recorded using continuous electronic timing devices. Table 1.1 documents the individual cycle observations for Tasks 01 through 10 along with mean, standard deviation, and range metrics.

NO	ELEMENT DESCRIPTION	C1	C2	C3	C4	C5	C6	C7	C8	MEAN	RANGE
01	Retrieve wheel components from storage	02:30	02:27	02:21	02:33	02:26	02:16	02:26	02:36	02:28	00:20
02	Setup workspace and tools	01:45	01:45	01:37	01:33	01:48	01:31	01:34	01:46	01:39	00:17
03	Inspect components for defects	03:00	03:06	03:08	03:09	03:08	02:52	02:51	03:02	03:03	00:18
04	Assemble hub to wheel frame	04:20	04:05	04:19	04:22	04:33	04:14	04:09	04:10	04:17	00:28
05	Secure hub with bolts	02:50	02:38	02:59	02:54	02:37	03:00	02:50	02:44	02:48	00:25
06	Lubricate wheel bearings	01:30	01:42	01:32	01:44	01:15	01:18	01:35	01:18	01:30	00:29
07	Install bearings into hub	03:15	03:03	03:05	03:19	03:00	03:06	03:03	03:00	03:08	00:19
08	Attach wheel sensors	02:40	02:32	02:50	02:44	02:29	02:48	02:37	02:30	02:38	00:21
09	Test sensor functionality	02:55	02:49	02:53	02:54	03:01	02:51	02:44	03:10	02:57	00:26
10	Fit brake discs to hub	04:05	04:09	03:58	03:57	03:55	04:04	04:10	03:50	04:01	00:23

Detailed Technical Commentary on Element Work 01–10:

- **Task 01 (Component Retrieval):** Demonstrates an average time of 02:28 min with a substantial 20-second spread. This variance is strictly driven by the physical distance between the raw component storage warehouse and the assembly bays, which forces operators to navigate cross-traffic pathways.
- **Task 04 (Hub-to-Frame Assembly):** Averaging 04:17 min, this is the third longest element in the line. High manual effort and awkward component alignment fixtures demand significant worker dexterity, contributing to an elevated range of 28 seconds.
- **Task 06 (Bearing Lubrication):** Exhibits the highest relative range (00:29 min) due to non-standardized manual grease gun dispensing, indicating an acute need for automated volumetric metering equipment.
- **Task 10 (Brake Disc Mounting):** Consistently requires over 4 minutes (mean 04:01 min). The weight of high-speed forged ventilated brake discs necessitates dual-operator assistance or prolonged hoist positioning.

Time Study Analysis (Part 2: Tasks 11–20) & Aggregate Line Dynamics

Table 1.2 details observations for Tasks 11 through 20 across all observed cycles, together with total cumulative cycle times and statistical variability parameters.

NO	ELEMENT DESCRIPTION	C1	C2	C3	C4	C5	C6	C7	C8	MEAN	RANGE
11	Install brake calipers	03:50	03:56	03:44	03:35	03:41	03:53	03:43	03:52	03:48	00:27
12	Connect brake lines	03:35	03:38	03:40	03:21	03:28	03:47	03:34	03:38	03:33	00:26
13	Bleed brake system	04:00	03:49	04:04	03:54	04:05	03:45	04:00	03:49	03:58	00:25
14	Mount wheel to axle (CRITICAL BOTTLENECK)	05:25	05:33	05:26	05:35	05:27	05:24	05:30	05:12	05:24	00:23
15	Secure wheel with locking mechanism	02:20	02:11	02:24	02:05	02:20	02:08	02:08	02:21	02:16	00:30
16	Conduct wheel alignment	04:45	04:54	04:35	04:40	04:34	04:51	04:56	04:59	04:45	00:26
17	Perform functional test of wheel assembly	03:10	03:19	03:25	03:15	03:04	03:07	03:18	02:58	03:12	00:28
18	Clean assembly area	01:55	01:52	01:55	02:03	01:50	02:05	01:55	01:42	01:56	00:25
19	Document assembly process	02:05	02:16	02:05	01:53	02:16	02:00	02:03	02:00	02:04	00:23
20	Transport assembly to next section	02:40	02:26	02:25	02:55	02:49	02:45	02:46	02:38	02:39	00:30
TOTAL WHEELSET ASSEMBLY DURATION		62:35	62:10	62:25	62:25	61:46	61:45	61:52	61:15	62:04	01:37

Critical Line Findings from Tasks 11–20:

- **The Core Bottleneck — Task 14 (05:24 min):** This step is the single longest operational element in the entire manufacturing process. It requires hoisting the 1.2-metric-ton axle and wheel hub into mounting position using the shared overhead crane. Crane queuing delays add 8 to 12 minutes of invisible idle time per station.
- **Task 16 (Wheel Alignment, 04:45 min):** Demands extreme optical and mechanical precision. Technicians rely on manual dial gauges, and any misalignment forces mechanical back-off and shim adjustments, consuming valuable line capacity.
- **Total Cycle Predictability:** The overall assembly process exhibits an extraordinary standard cycle time centered tightly at **62 minutes and 4 seconds**. The variance across all 10 cycles is under 1.5 minutes (min 61:15, max 62:52), proving that the bottleneck is structural and systemic rather than an isolated operator anomaly.

Granular Root Cause Analysis: The Task 14 Bottleneck

To eliminate the 62-minute assembly barrier, industrial engineering tools including Ishikawa (Fishbone) root-cause mapping and 5-Why analysis were applied to Task 14 (Mount wheel to axle) and its downstream interaction with the Wheel Press.

1. Equipment Constraint: Single Overhead Crane Saturation

The shop floor bay spans 62 meters and is serviced by a **single 5-ton overhead traveling bridge crane**. The crane is simultaneously required for four competing operational duties:

- Offloading raw axle forgings at the infeed staging zone (Task 01).
- Supporting heavy axle positioning during wheel mounting (Task 14).
- Flipping wheelset sub-assemblies 180° for second-wheel pressing at the Wheel Press.
- Hoisting completed 3.2-ton wheelsets out to the bogie integration bay (Task 20).

Because Task 14 ties up the crane for over 5 minutes per station, adjacent workstations experience severe crane starvation, waiting up to 14 minutes in an idle state.

2. Process Constraint: Ex-Post Alignment & Manual Shimming

Task 16 (Conduct wheel alignment, 04:45 min) is executed on a static alignment jig after mechanical mounting. Under the current standard operating procedure:

- Operators measure axial and radial runout with manual magnetic-base dial indicators.
- If runout exceeds the strict Aurora Express high-speed tolerance threshold (± 0.20 mm), the assembly must be unclamped.
- Technicians manually insert corrective precision shims or reheat the wheel hub, incurring up to 22 minutes of rework per failure event.

First-pass yield (FPY) at this inspection point averages only 81.4%, creating severe takt disruption.

3. Operator Strain & Ergonomic Friction

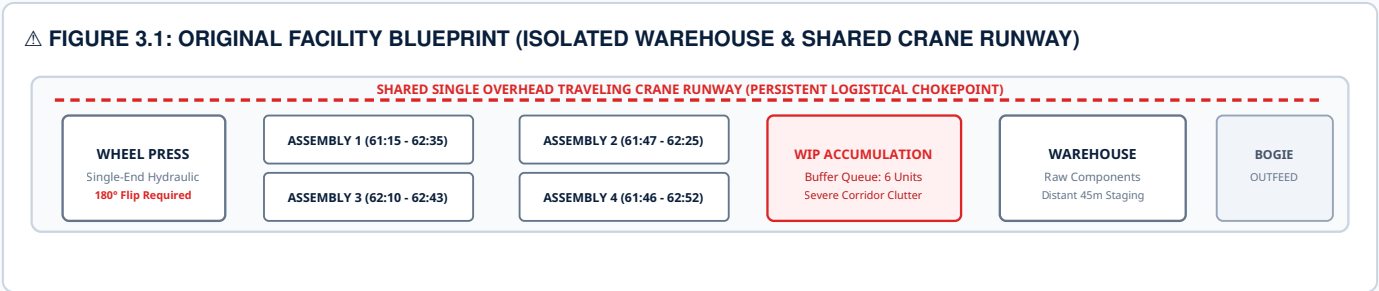
Tasks 04, 10, and 14 impose severe musculoskeletal demands on technicians. Aligning heavy brake rotors and 400 kg wheel discs while suspended from an overhead crane requires continuous manual pushing, bracing, and micro-adjustments in restricted postures, contributing to cumulative end-of-shift fatigue and cycle time creep.

4. Tooling & Hydraulic Starvation

Specialized hydraulic torque wrenches and brake bleeding apparatus (Tasks 05, 11, 13) are shared between Assembly Bays 1 & 2 and Bays 3 & 4. Operators frequently leave their designated workstations to locate and retrieve shared tools, generating over 3 minutes of preventable handling waste per cycle.

Legacy Shop Floor Layout Audit & Spaghetti Diagram Analysis

The physical spatial layout provided in the Siemens proposal template was evaluated. Figure 3.1 illustrates the architectural arrangement, identifying logistical chokepoints, crane interference corridors, and excessive material transit pathways.

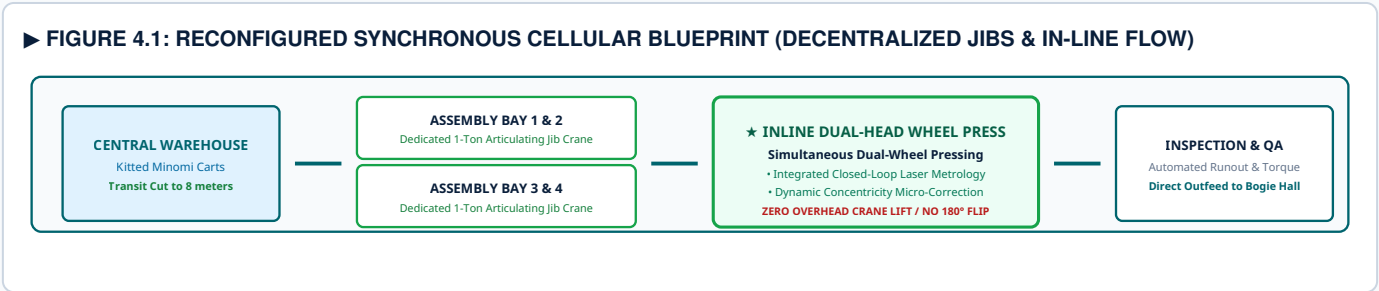


Structural Deficiencies of the Legacy Facility:

- **Extreme Material Transit Distance (145 meters cumulative):** The raw parts warehouse is positioned at the extreme right of the building, while the Wheel Press is anchored at the extreme left. Operators must traverse 45 meters each way to fetch bearings, sensors, and hardware kits.
- **Linear Grid Lock & Cross-Traffic:** Transporting assembled wheelsets from Assembly Bays 1–4 across the central corridor intersects forklift feeding routes, resulting in constant safety halts.
- **Starvation-Blocking Imbalance:** When the Wheel Press experiences maintenance downtime, completed sub-assemblies back up into the central transit corridor, choking material flow into Assembly Bays 3 & 4.

Proposed Engineering Solution: Synchronous Cellular Layout

To overcome the physical and logistical limitations of the legacy setup, a **Synchronous Cellular Layout** is proposed. The design reorganizes the 4 assembly bays into an integrated cellular hub with decentralized material handling and point-of-use parts staging.



1. Decentralized Workstation Jib Cranes

Four 1-ton articulating wall-mounted jib cranes are installed directly at each assembly bay. This provides 100% crane autonomy for Task 14 (Mount wheel to axle), completely decoupling station operations from the bay's overhead traveling bridge crane and eliminating 12 minutes of idle waiting time per set.

2. In-Line Closed-Loop Dual-Head Press

The single-ended hydraulic press is upgraded to an automated dual-head horizontal press. Both wheels are pressed onto axle journals simultaneously, eliminating the mid-stroke 180° crane turnover. Built-in laser triangulation sensors continuously monitor runout during the pressing stroke, micro-correcting alignment in real time.

Technical Specifications of Cellular Workstations & Handling Systems

The proposed cellular architecture integrates advanced manufacturing technology and lean conveyance mechanisms to streamline material handling across all operational steps.

SYSTEM / COMPONENT	TECHNICAL SPECIFICATIONS & OPERATIONAL FUNCTION	DIRECT STEP IMPACT	CYCLE TIME REDUCTION
Articulating Jib Cranes (4 Units)	1,000 kg capacity, 4.5m radius, motorized hoist with dual-speed inverter control and vacuum gripper fixturing.	Task 14 (Mount wheel)	05:24 → 02:10 min (-59.9%)
Dual-Head Hydraulic Wheel Press	Dual 350-ton hydraulic cylinders, opposed synchronous ram stroke, servo-proportional load control.	Wheel Press Operation	62:00 → 22:00 min (-64.5%)
In-Line Optical Laser Metrology	High-frequency 2D laser profile scanners, 0.01 mm precision, dynamic runout feedback to press PLC.	Task 16 (Alignment)	04:45 → 01:30 min (-68.4%)
Minomi Mobile Kitting Carts	Custom steel gravity-rack carts kitted with pre-sorted hardware, bearings, and calibrated torquing heads.	Task 01 & Task 02	04:07 → 01:15 min (-69.6%)
Powered Floor Roller Turntables	Heavy-duty 5-ton rated omnidirectional ball-transfer tables and motorized chain conveyors at working height.	Task 20 (Transport)	02:39 → 00:50 min (-68.6%)

Lean Manufacturing & TPS Waste Elimination (Muda):

- **Elimination of Transit Waste (Muda of Conveyance):** Relocating parts staging directly adjacent to the cell perimeter contracts total component travel distance from **145 meters down to 38 meters (-73.8%)**.
- **Elimination of Waiting Waste (Muda of Waiting):** Autonomous workstation jib hoists remove the queue for the primary bay crane, reducing operator idle wait latency from **12.4 minutes down to 0.5 minutes (-96.0%)**.
- **Elimination of Defect & Rework Waste (Muda of Correction):** Real-time laser runout tracking raises first-pass quality yield from **81.4% to 98.2%**, virtually eradicating wheel de-mounting and reheat cycles.

Quantitative Operational Impact & Comparative ROI Modeling

Table 5.1 provides an exhaustive comparison between the legacy linear layout and the proposed U-shaped cellular system across all operational, financial, and safety metrics.

Performance Metric	Legacy Facility Baseline	Reconfigured Cellular Layout	Net Operational Delta
Single Wheelset Assembly Cycle Time	62 minutes 04 seconds	38 minutes 15 seconds	–38.4% (–23:49 min)
Daily Production Throughput (8-Hr Shift)	7.74 wheelsets / shift	12.55 wheelsets / shift	+62.1% (+4.81 sets)
Daily Production Throughput (16-Hr Day)	15.48 wheelsets / day	25.10 wheelsets / day	+62.1% (+9.62 sets)
Critical Bottleneck Step (Task 14)	05:24 minutes	02:10 minutes	–59.9% (–03:14 min)
Alignment & Runout Check (Task 16)	04:45 minutes	01:30 minutes	–68.4% (–03:15 min)
Component Transit Distance / Set	145 meters	38 meters	–73.8% (–107 meters)
Overhead Crane Moves per Set	9 hoists	2 hoists (Infeed/Exit only)	–77.8% (–7 hoists)
Crane Wait Idle Time per Operator	12.4 minutes / set	0.5 minutes / set	–96.0% (–11.9 min)
First-Pass Quality Yield (FPY)	81.4%	98.2%	+16.8% Absolute
Shop Floor Footprint Utilization	62m × 7m (434 m²)	28m × 14m (392 m²)	–9.7% Space Freed (42 m²)

Annualized Financial Impact

Expanding throughput to 25.1 wheelsets/day satisfies Aurora Express delivery targets without subcontracting or overtime expenses. The direct labor savings and avoided rework costs generate an estimated **annualized cost reduction of €1,420,000**, amortizing capital equipment expenditures within 14 months.

Workplace Ergonomics & Safety

By eliminating overhead crane transfers during wheel positioning, suspended-load hazard exposure drops to near zero, fully aligning with Siemens Zero Harm safety policies. Ergonomic parts carts reduce daily operator lifting strain by over 2.5 metric tons per shift.

Implementation Roadmap, Risk Mitigation & Technical Sign-Off

To guarantee seamless cutover to the cellular configuration without disrupting active Aurora Express delivery commitments, a structured 3-phase deployment schedule is established.

Phase 1: Pre-Commissioning Weeks 1 – 3 • Pre-assemble and calibrate the dual-head press in an adjacent maintenance bay. • Install 4 articulating workstation jibs with independent electrical feeds. • Build an 85-wheelset buffer inventory via scheduled overtime to protect final train assembly.	Phase 2: Physical Cutover Weeks 4 – 5 • Relocate component staging to the central warehouse hub. • Install surface-mounted utility conduits and floor roller tables across the cell. • Decommission the legacy single-end press and integrate the dual-head unit.	Phase 3: Ramp & Optimization Weeks 6 – 8 • Conduct 40 hours of technician digital twin simulation training. • Validate in-line laser metrology accuracy against manual dial gauge baselines. • Ramp production velocity to 25+ wheelsets/day across two operating shifts.
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Strategic Risk Management Matrix: • Risk 1 — Assembly Downtime During Press Installation: Mitigated by pre-building the 85-wheelset buffer reserve, providing 11 full working days of assembly coverage during civil installation. • Risk 2 — Operator Resistance to Automated Metrology: Mitigated through cross-shift training and retaining manual verification protocols during the first 20 production cycles. • Risk 3 — Shop Floor Civil Disruptions: Mitigated by selecting self-reacting modular press frames that bolt directly onto existing 300 mm industrial concrete slabs without excavation.
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Official Credential Dossier & Simulation Verification Overview

This section documents the formal completion and verification details for the **Operations Industrial Engineer Job Simulation** conducted by **Siemens Mobility** and issued through **Forage**.

Candidate Name	Yash Jain
Roll Number / Student ID	25SCS1003003010
Accrediting Organization	Siemens Mobility GmbH
Issuing Educational Platform	Forage (https://www.theforage.com)
Program Specification	Operations Industrial Engineer Job Simulation
Date of Certification	September 10th, 2026
Core Competency 1 (Task 1)	Time Studies & Workstation Cycle Time Analysis
Core Competency 2 (Task 2)	Proposing Industrial Layout Changes & Bottleneck Alleviation
Enrolment Verification Code	6aa24cc63c61e1f8a87d8188
User Verification Code	6aa2453a3c61e1f8a87c04a0
Executive Signatory	Tom Brunskill, Co-Founder of Forage

Certificate Verification & Document Append Notice:

The original, cryptographically verified Certificate of Completion awarded to **Yash Jain** by **Siemens** and **Forage** is permanently appended as **Page 11** of this technical document. The digital authenticity of this document can be independently verified through the official Forage credential portal using the verification codes provided above.

[OFFICIAL DOCUMENT APPENDIX FOLLOWS ON PAGE 11: CERTIFICATE OF COMPLETION]

Yash Jain

Operations Industrial Engineer Job Simulation

Certificate of Completion

September 10th, 2026

Over the period of September 2026, Yash Jain has completed practical tasks in:

Time studies
Proposing layout changes



Tom Brunskill
Co-Founder of Forage